FINAL YEAR RESEARCH PROJECT

End-term Presentation

An Explainable AI-based Loan Approval Prediction System for Transparent Credit Decision Making

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Presentation Outline

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  - Motivations
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Problem Statement

Loan approval in traditional systems is often time-consuming and lacks transparency. Applicants are not provided with clear reasons for approval or rejection. This creates a need for an intelligent system that can predict loan approval and explain the decision.

Motivations

- Traditional loan approval processes are slow and involve manual verification.
- Human decision-making may lead to inconsistency or bias.
- Machine learning can automate decision-making using historical data.
- Explainable AI helps improve transparency in prediction.
- User interface enables real-time practical use of the model.

Literature Review

Table 1:Summary of Existing Approaches and Research Limitations

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
1)	Sinap (2024)	Logistic Regression, KNN, SVM, Decision Tree, Random Forest, RFE, K-Best	Loan approval dataset (not clearly specified)	Compared ML models with feature selection methods	Random Forest achieved 97.71% accuracy; RFE improved performance significantly
2)	Haque & Hassan (2023)	AdaBoost, Random Forest, SVM, Decision Tree, GaussianNB	148,670 records, 37 features	Built predictive model for loan approval	AdaBoost achieved highest accuracy (99.99%), showing strong ensemble performance
3)	Literature Review Studies	Ensemble models, Logistic Regression, Random Forest, SVM	Various datasets (Kaggle, bank data)	Compared multiple ML approaches	Ensemble methods improve accuracy; model performance varies by dataset
4)	General ML Loan Systems	Automated ML-based loan approval	Financial & banking datasets	Replaced manual loan approval processes	Faster, efficient, but depends on data quality and model selection
5)	Explainable AI Research	SHAP, LIME	LendingClub dataset		Provides explanation for model decisions, improves transparency

Literature Review

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
6)	Ayad et al. (2024)	Explainable AI, SHAP, ML Classification Models	“Give Me Some Credit” dataset (120K+ records)	Developed explainable loan approval prediction system	XAI improved transparency and helped bankers understand approval decisions clearly
7)	Li et al. (2024)	Random Forest, XGBoost, SHAP, SVM, KNN	Lending institution loan dataset	Compared 9 ML models for loan default prediction	Random Forest showed best stability and SHAP identified important risk factors like age and employment years
8)	Nallakaruppan et al. (2024)	Explainable AI Framework, Credit Evaluation Models	Financial credit datasets	Built recommendation and support system for banks	XAI improved decision support, reduced false positives, and increased customer transparency
9)	Xie et al. (2025)	Logistic Regression, Random Forest, XGBoost, SHAP, SMOTE	Automobile loan default dataset	Improved loan default prediction using explainable feature selection	SHAP + SMOTE-Tomek achieved superior prediction performance with better handling of imbalanced data
10)	Pathak & Shreya (2025)	XGBoost, LightGBM, Random Forest, SHAP, LIME	Credit Risk Assessment dataset	Developed transparent AI-based credit risk assessment system	LightGBM achieved best balance between prediction accuracy and explainability using SHAP/LIME reports

Research Gap & Improvements

Research Gap

- Lack of transparency in existing loan approval systems.
- Limited integration of explainable AI with multiple ML models.
- Black-box decision-making reduces user trust.
- Limited real-time and user-friendly implementations.

Improvements

- Integrated ML with Explainable AI (SHAP).
- Compared multiple ML algorithms.
- Added transparent prediction explanations.
- Developed real-time Streamlit-based interface.

Architecture

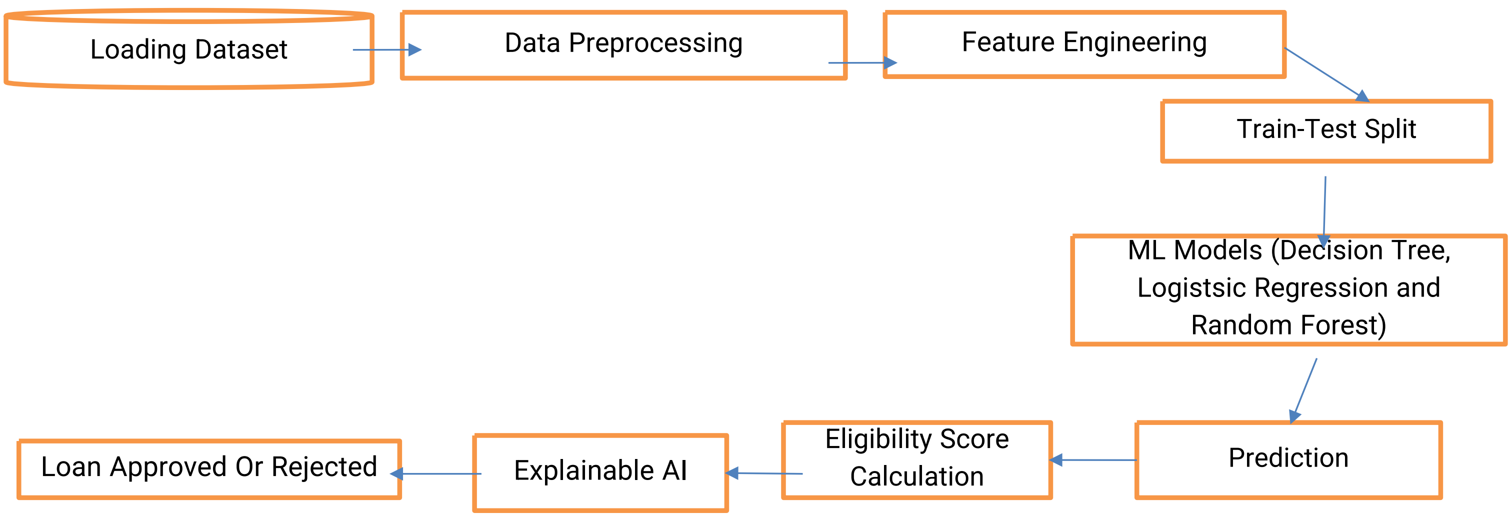


Fig. 1: Workflow of the Proposed Model

Architecture

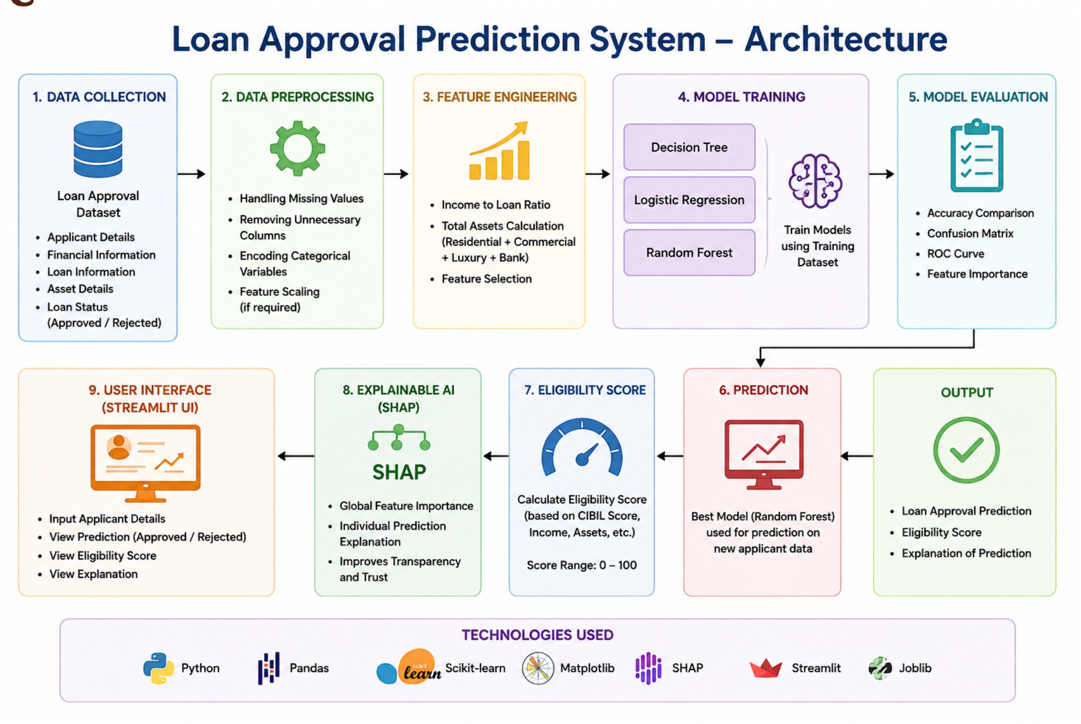


Fig. 2: Schematic Model Diagram of the Proposed Model

Proposed Solution

- Develop an intelligent loan approval prediction system using machine learning.
- Analyze applicant financial details such as income, CIBIL score, assets, and loan amount.
- Train multiple models to predict loan approval status accurately.
- Select the best-performing model for final deployment.
- Provide real-time prediction and eligibility score through a user-friendly Streamlit interface.

Dataset

Loan Approval Dataset (Kaggle)

It contains features such as:

- ❑ Income
- ❑ Loan Amount
- ❑ CIBIL Score
- ❑ Employment Status
- ❑ Asset Values
- ❑ Loan Status (Target Variable)

Tools And Technologies Used

- Python
- Jupyter Notebook
- Pandas & NumPy (Data handling)
- Matplotlib & Seaborn (Visualization)
- Scikit-learn (Machine Learning models)

Experimentation

- Data Preprocessing
- Removed missing and duplicate values
- Encoded categorical features using Label Encoding
- Normalized numerical features for better model performance
- Split dataset into Training (80%) and Testing (20%) sets
- Machine Learning Models Used
- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- XGBoost Classifier
- K-Nearest Neighbors (KNN)
- Model Evaluation Metrics
- Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, Confusion Matrix
- Explainable AI
- SHAP used for feature importance analysis
- Generated transparent explanations for predictions

Results And Analysis

Experimental Results

• Random Forest and XGBoost achieved highest prediction accuracy
• Logistic Regression provided stable and interpretable results
• Ensemble models performed better than traditional models
• SHAP analysis identified CIBIL score, income, and loan amount as major influencing factors

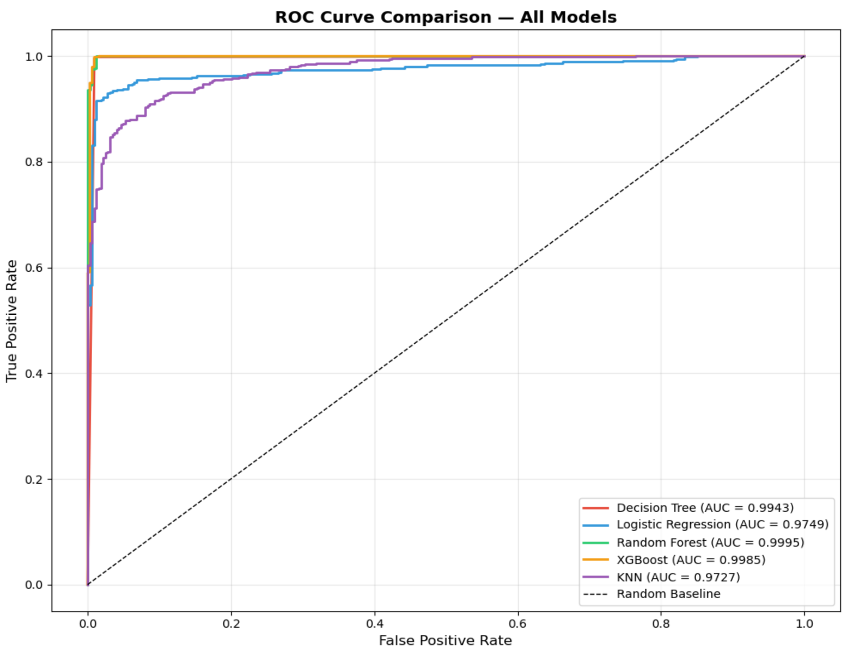
Performance Analysis

• Cross-validation improved model reliability
• ROC-AUC curves showed strong classification capability
• Confusion matrices indicated low false prediction rates
• Explainable AI increased transparency and trust in predictions

Key Observation

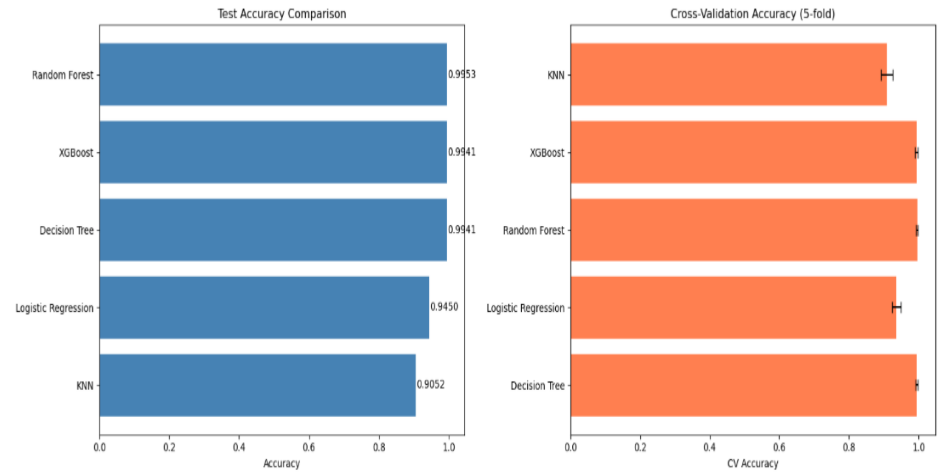
• Higher CIBIL score and stable income significantly improve loan approval probability

Model Performance Comparison



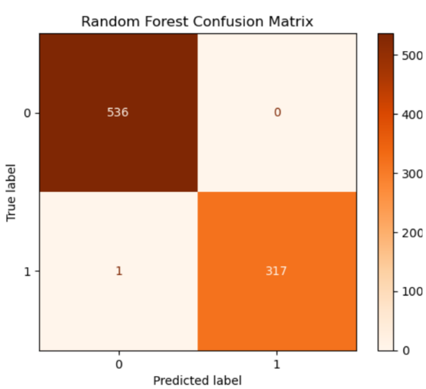
- Multiple machine learning models were trained and evaluated.
- Random Forest and XGBoost achieved the highest accuracy.
- Ensemble models outperformed traditional classifiers.

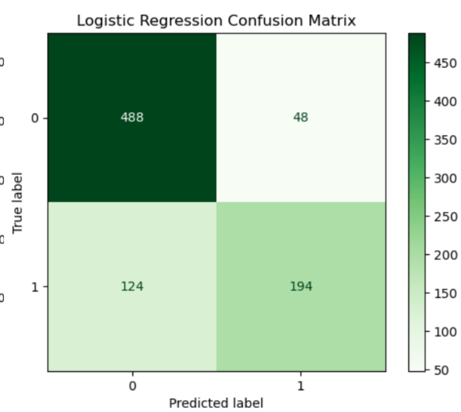
Model	Test Accuracy	Precision	Recall	F1 Score	AUC-ROC	CV Accuracy (mean)	CV Accuracy (std)
Decision Tree	0.9941	0.9925	0.9981	0.9953	0.9943	0.9953	0.0033
Logistic Regression	0.9450	0.9654	0.9454	0.9553	0.9749	0.9367	0.0124
Random Forest	0.9953	0.9925	1.0000	0.9962	0.9995	0.9959	0.0028
XGBoost	0.9941	0.9925	0.9981	0.9953	0.9985	0.9944	0.0043
KNN	0.9052	0.9061	0.9454	0.9253	0.9727	0.9092	0.0171

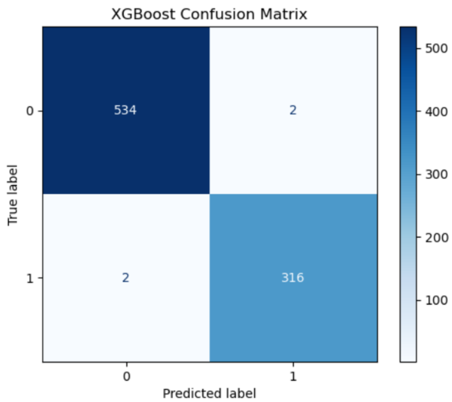


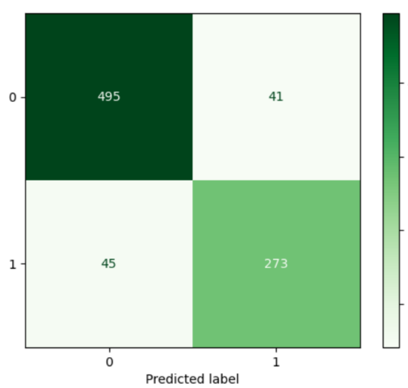
Performance metrics were analyzed using Accuracy, Precision, Recall, and F1-score.

Confusion Matrixes



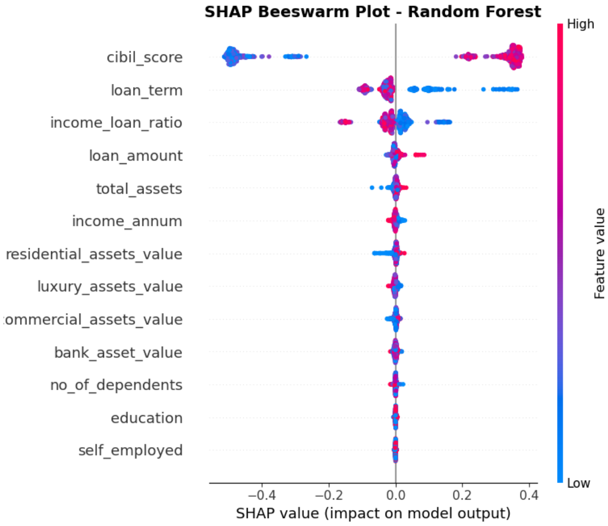


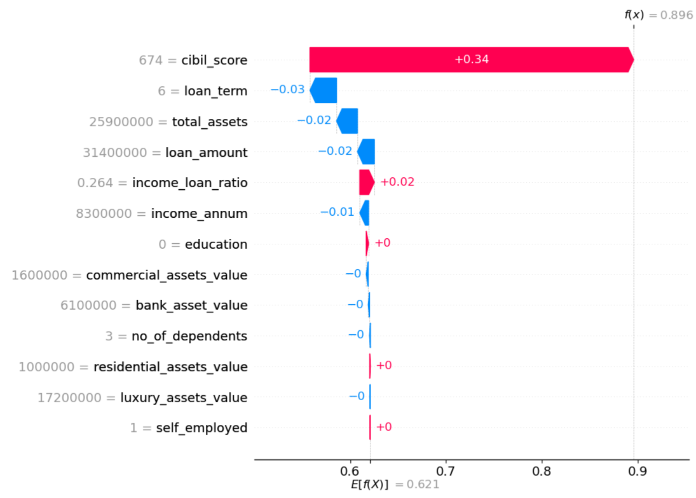




- Confusion matrices were used to evaluate classification performance.
- Random Forest produced the lowest misclassification error.
- Logistic Regression showed balanced prediction capability.
- XGBoost handled classification efficiently with fewer false predictions.

SHAP Explainability Analysis



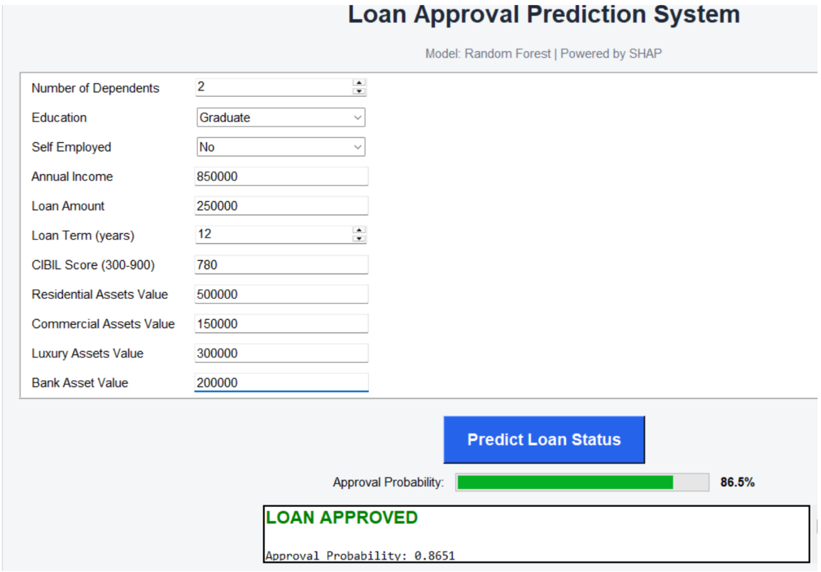


- SHAP was used to explain prediction decisions.
- CIBIL score, income, and loan amount were identified as the most influential features.
- Explainable AI improved transparency and interpretability of the system.
- Feature contribution analysis increased trust in automated loan approval decisions.

Observation

• Higher CIBIL score and stable income positively influence loan approval probability.

Final System Output



- A user-friendly Tkinter-based graphical interface was developed for real-time loan approval prediction.
- Users can enter applicant details such as income, CIBIL score, assets, and loan amount.
- The system predicts whether the loan will be approved or rejected.
- The interface also displays eligibility score and prediction analysis.
- The GUI improves usability and enables practical implementation of the proposed model.

Conclusion

Developed an Explainable AI-based loan approval prediction system

Compared multiple machine learning algorithms for prediction accuracy

Random Forest/XGBoost showed superior performance

SHAP improved transparency by explaining prediction outcomes

Streamlit interface enabled real-time user interaction

Proposed system can support faster and smarter loan approval decisions

Future Scope

1. Use larger real-world banking datasets
2. Improve performance using Deep Learning techniques
3. Integrate fraud detection and risk analysis modules
4. Deploy system on cloud platforms for real-time banking applications
5. Enhance explainability using advanced XAI techniques like LIME and Counterfactual AI
6. Add multilingual and mobile-friendly user interface

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Thank you